A Stylometric Application of Large Language Models

Harrison F. Stropkay , Jiayi Chen , Mohammad J. Latifi , Daniel N. Rockmore , Jeremy R. Manning

Dartmouth College, Hanover, NH 03755, USA

May 2026

Source: arXiv:2510.21958 • Code: https://github.com/ContextLab/llm-stylometry

We show that large language models (LLMs) can be used to distinguish the writings of different authors. Specifically, an individual GPT-2 model, trained from scratch on the works of one author, will predict held-out text from that author more accurately than held-out text from other authors. We suggest that, in this way, a model trained on one author's works embodies the unique writing style of that author. We first demonstrate our approach on books written by eight different (known) authors. We also use this approach to confirm R. P. Thompson's authorship of the well-studied 15th book of the Oz series, originally attributed to F. L. Baum.

1 Introduction

Herein we introduce *predictive comparison*, a new LLM-based relative stylometric measure. It derives from a simple idea, that if an LLM can be trained to write like a given author by training on their work, then the degree to which such a model can predict held-out text from that author should reflect the strength of the author's style.

Stylometry has a long history in computational linguistics. Classical work relies on lexical, syntactic, and structural features—word frequencies, function-word distributions, part-of-speech *n*-grams. Modern neural methods replace hand-crafted features with learned representations.

2 Approach

For each candidate author A in a set, we train a small GPT-2 model on a training partition of A's writings, then evaluate cross-entropy on held-out passages from every author in the set.

- Content tokens: the full token stream.
- Function words: content words masked out.
- POS tags: every token replaced by its part-of-speech.

We use Python and PyTorch; the full training and evaluation code is at ContextLab/llm-stylometry.

2.1 Cross-entropy

For an author A trained model θ_A and a held-out passage x from author B, the per-token cross-entropy is

$$H(x; \theta_A) = -\frac{1}{|x|} \sum_{t=1}^{|x|} \log p_{\theta_A}(x_t \mid x_{<t}).$$

The lower the cross-entropy, the more author-B-like the model finds the passage.

2.1.1 Statistical comparison

We compare cross-entropies via two-sample *t*-tests over passages.

Discussion

A small accent rule, a colored section number, a few links: that's what we need to look at to verify the house style is doing what we want.

Code availability

Code: `https://github.com/ContextLab/llm-stylometry`.